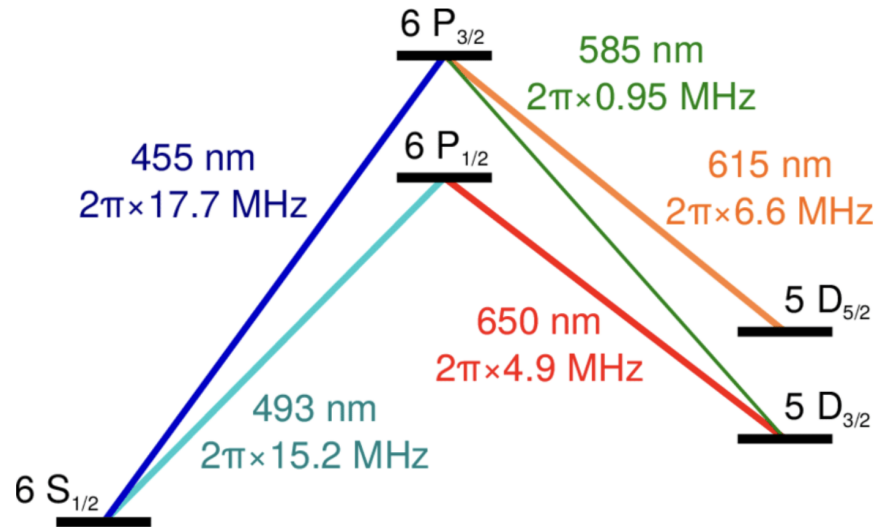


Simulations of Ba⁺ Ion Qubit coherence time

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2: Level scheme of Barium with relevant transitions for laser cooling and repumping. The main transitions relevant for Doppler cooling are the 493 nm (S 1/2 → P 1/2) and 650 nm (D 3/2 → P 1/2) transitions. However, off-resonant scatter

Two Level Structure of Ba⁺

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1 Introduction

Coherent control of quantum two-level systems is one of the most important needs of modern quantum technologies, including trapped-ion quantum computation [1], precision spectroscopy, and quantum sensing. When a two-level atom is driven by a near-resonant electromagnetic field, coherent Rabi oscillations enable controlled population transfer and state manipulation. In realistic experimental settings, however, imperfections such as laser frequency and amplitude lead to dephasing and a loss of coherence, thereby limiting the fidelity of quantum operations. Developing methods to understand and mitigate these effects is therefore of fundamental and practical importance.

In this project, we investigate the dynamics of a driven two-level system (TLS) using both analytical and numerical approaches, with particular emphasis on noise-induced decoherence and its suppression. The dressed-state and double-dressed-state [2] formalisms are employed to gain physical insight into the system's behavior under strong and time-dependent driving fields. Using the optical Bloch equations (OBE's) we study the effects of static and time-dependent detuning as well as Rabi-frequency fluctuations on coherent dynamics. Building on recent theoretical proposals, we further analyze phase-modulated driving as a form of continuous dynamical decoupling [3, 4], demonstrating how the introduction of an additional modulated field can effectively average out noise and extend the coherence time of the system. The results can help get a clear physical interpretation of noise suppression mechanisms and are directly relevant to experimental implementations in trapped-ion systems.

2 Theory

2.1 Solving Open Quantum Systems

Throughout this report, the dynamics of the two-level system (TLS) are primarily analyzed using the optical Bloch equations (OBE's). Nevertheless, we briefly revisit alternative theoretical approaches and clarify the motivation for adopting this particular framework. Below, we outline other methods that can be employed to describe the dynamics of open quantum systems and discuss their relevance in the present context.

Equation type	Form	System type	Description
Schrödinger	$i\hbar \dot{\psi} = H\psi$	Single pure, isolated system	Governs the unitary time evolution of a pure quantum state.
Liouville–von Neumann	$\dot{\rho} = -\frac{i}{\hbar}[H, \rho]$	Isolated system (density-matrix form)	Equivalent to the Schrödinger equation, expressed in density-matrix formalism.
Master equation	$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}[\rho]$	Open system (coupled to environment)	Describes dissipative dynamics; $\mathcal{L}[\rho]$ is the Lindblad superoperator incorporating decay and dephasing.
Optical Bloch equations	Component form of master equation	Two-level driven atom (with or without damping)	Express the time evolution of populations and coherences in the Bloch-vector representation.

Table 1: Hierarchy of quantum dynamical equations from isolated to open systems.

Writing the master equation in terms of the density-matrix elements

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix}, \quad \rho_{ee} + \rho_{gg} = 1,$$

one obtains the **Optical Bloch Equations** for a driven two-level system with spontaneous emission:

$$\begin{aligned} \dot{\rho}_{ee} &= -\gamma \rho_{ee} + \frac{i\Omega}{2}(\rho_{ge} - \rho_{eg}), \\ \dot{\rho}_{eg} &= \left(i\Delta - \frac{\gamma}{2}\right) \rho_{eg} + \frac{i\Omega}{2}(2\rho_{ee} - 1), \\ \rho_{ge} &= \rho_{eg}^*. \end{aligned} \quad (1)$$

The parameter Ω is the *Rabi frequency*, which quantifies the strength of the coherent coupling between the two levels induced by the driving field. It sets the characteristic timescale of coherent population oscillations. The parameter γ is the *spontaneous emission rate* from the excited state, describing irreversible population decay and the associated loss of coherence. As one tries to solve these equations for any atom at point $\mathbf{r}$ for steady state solutions, one would notice a solution of the form of $1 - \cos^2(\frac{\Omega t}{2})$ or $\sin^2(\frac{\Omega t}{2})$ which means that the population dynamics between the ground and excited state is performing an oscillations; often referred as the *rabi oscillation*.

2.2 Dressed Atom Approach

The dressed atom approach tackles the atom field coupling in a different way; the combined coupled hamiltonian (atom + laser photons) is diagonalized to obtain the so called *dressed states*, further then the coupling of dressed atom with the empty modes of the electromagnetic field. Consider a dressed hamiltonian depending on position $\vec{r}$ is given by [2];

$$\mathcal{H}_{DA}(\vec{r}) = \hbar(\omega_L - \delta)b^\dagger b + \hbar\omega_L a_L^\dagger a_L - [\mathbf{d} \cdot \mathcal{E}_L(\mathbf{r})b^\dagger a_L + \mathbf{d} \cdot \mathcal{E}_L^*(\mathbf{r})ba_L^\dagger] \quad (2)$$

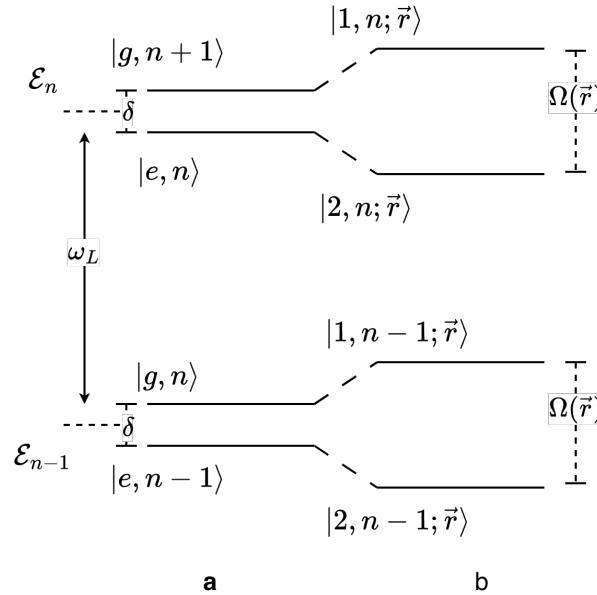


Figure 1: Dressed atom energy diagram, a: States of the combined atom laser mode without coupling, b: Dressed states at given point $\vec{r}$

If the atom laser mode coupling is taken into account ($\mathbf{d} = 0$), the eigenstates of the dressed hamiltonian are bunched in manifolds $\mathcal{E}_n$; n is a integer, separated by energy $\hbar\omega_L$ each manifold consisting of the two states $|g, n+1\rangle$ and $|e, n\rangle$ (atom in the internal state g or e in the presence of $n+1$ or n laser photons). The atom-laser coupling connects only the two states of a given manifold transition from g to e with the absorption of one laser photon. Shown is dressed atom energy diagram fig. 1. This dressed atom approach can be often referred as dressed basis for the rest of the report.

3 Simulations & Results

Simulations were performed with the aim of realizing different physical decoupling schemes. Shown below are the different simulations ran for this project, in code implementing these different work on same principle of solving the OBE written as eq 1.

3.1 Rabi Oscillations

In these simulations, we first solved the optical Bloch equations (OBEs) and plotted the resulting Rabi oscillations. We then investigated how these oscillations behave in the dressed-state basis to determine whether the characteristic Rabi dynamics are preserved. Here the Rabi Frequency of the system is denoted by Ω , Detuning by δ and Damping factor by γ (due to spontaneous emission).

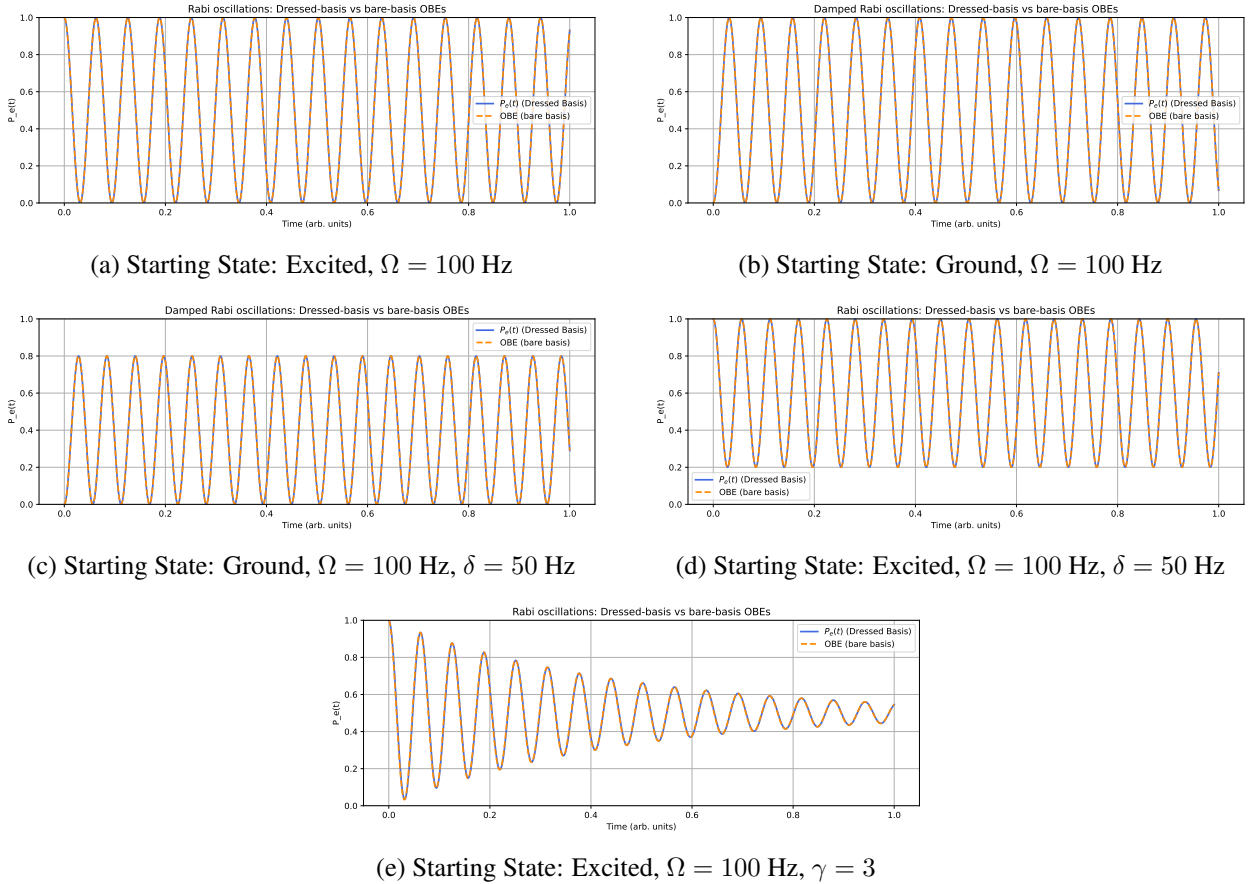


Figure 2: Rabi oscillations under various initial conditions and parameters.

3.2 Rabi Oscillations & Dephasing

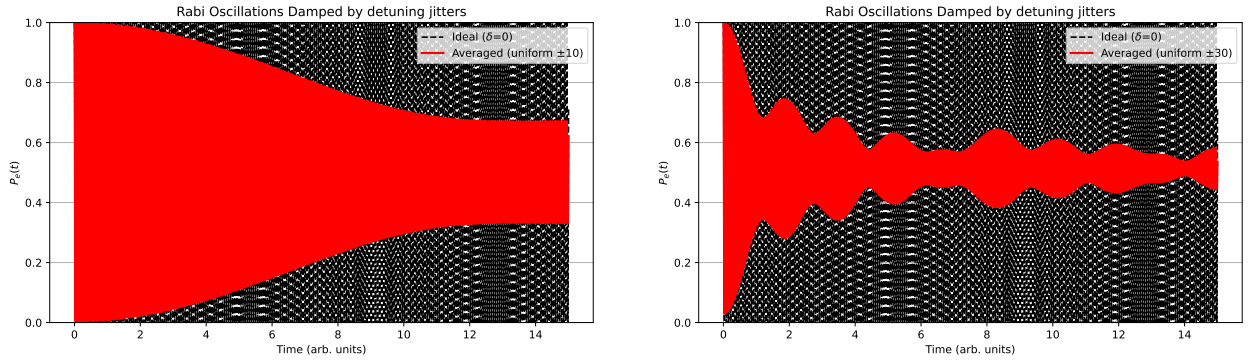
In these simulations, we aimed to move one step closer to realistic experimental conditions. In practice, small fluctuations in the detuning

$$\delta = \omega_L - \omega_0$$

are unavoidable. While such minute jitters have a negligible effect in a single experimental run, their impact becomes significant when many repetitions are performed for statistical averaging, ultimately leading to dephasing. This behavior is also reflected in the Rabi oscillations observed in our simulations. To model detuning jitters, we allowed the detuning to vary within a finite interval, for example

$$\delta \in [-10, 10],$$

which mimics experimental jitter. For each iteration, a random detuning value was drawn from this range and a full Rabi oscillation sweep was performed. Upon averaging (200 runs) over the resulting ensemble of trajectories, the cumulative effect of detuning fluctuations becomes clearly visible as a decay of the coherent oscillations.



(a) Starting State: Excited, $\Omega = 100$ Hz, $\delta \in [-10, 10]$,

(b) Starting State: Ground, $\Omega = 100$ Hz, $\delta \in [-30, 30]$

Figure 3: Rabi oscillations under various detuning ranges and averaged

3.3 Dephasing & Faster Drive

3.3.1 Time Independent Detuning

One way to mitigate the effects of detuning jitter ($\delta = \delta \pm \Delta\delta$) is to drive the system at a higher Rabi frequency, thereby making the coherent dynamics faster than the timescale of the fluctuations. This behavior can be readily visualized using the same simulation framework as before, in which the detuning is allowed to vary within a finite interval, for example

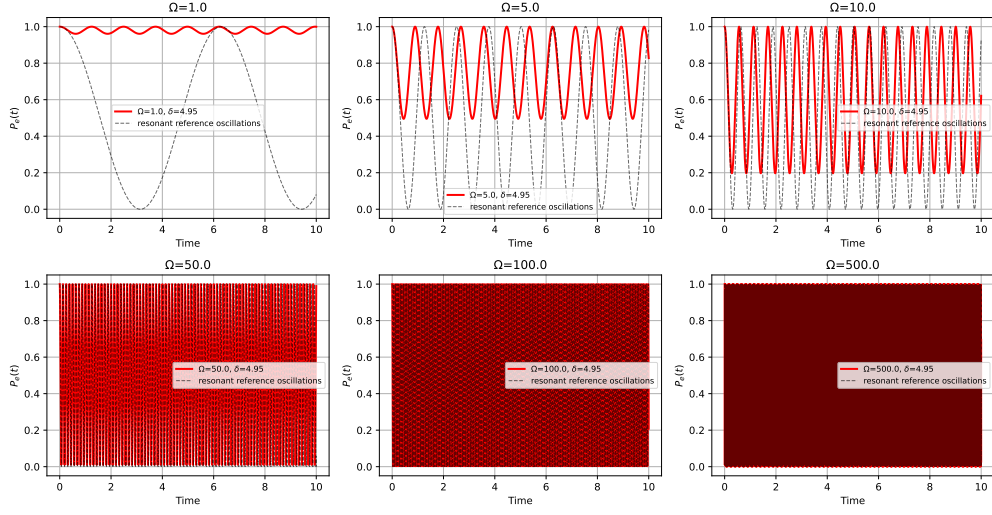
$$\delta \in [-10, 10].$$

For a fixed random value of the detuning drawn from this interval, we examined the resulting Rabi oscillations for different choices of the Rabi frequency. The results are summarized in Fig. 4. It is evident that, in the dressed-state basis and in the regime

$$\delta \ll \Omega,$$

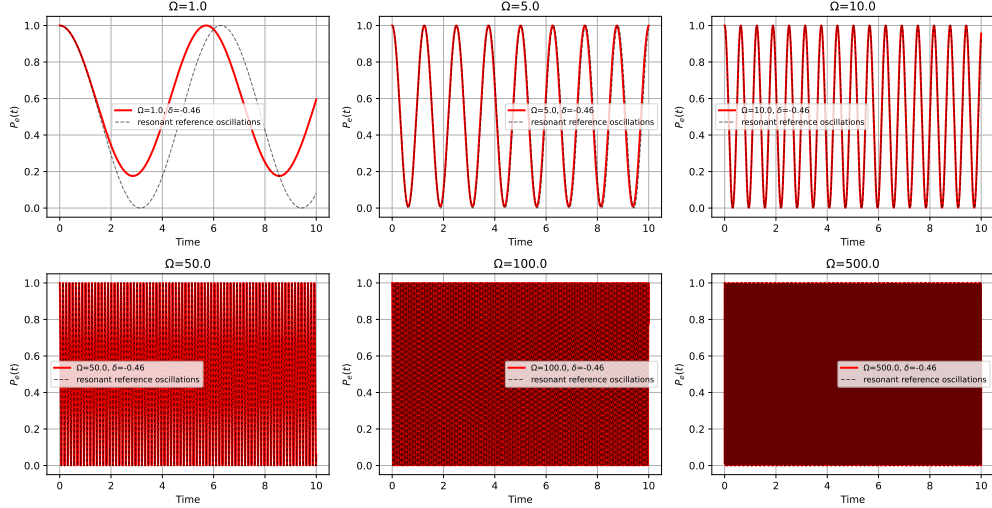
the effective two-level system becomes increasingly robust against detuning-induced jitter, with the coherent Rabi oscillations largely preserved. We test this with Rabi oscillations for different values of $\Omega \in [1, 5, 10, 50, 100, 500]$ for the same detuning.

Rabi oscillations for varying Ω with same $\delta \in [-10, 10]$



(a) Starting State: Excited, $\delta \in [-5, 5], \delta = 4.95$

Rabi oscillations for varying Ω with same $\delta \in [-10, 10]$



(b) Starting State: Ground, $\delta \in [-5, 5], \delta = 0.46$

Figure 4: Time Independent Detuning Simulations

3.3.2 Time dependent Detuning

A faster drive does indeed mitigate the effects of detuning jitter; however, the simulations discussed so far assumed a time-independent detuning, meaning that the detuning remains fixed during an individual experimental run. This assumption does not fully reflect realistic experimental conditions, where detuning fluctuations are inherently time dependent and can vary within a single run.

To address this, we extend our simulations to include time-dependent detuning jitters, thereby mimicking a more realistic experimental scenario. We show that even in the presence of such intra-run detuning fluctuations, increasing the Rabi frequency continues to enhance the robustness of the system, preserving coherent dynamics despite the jitter.

We therefore perform simulations in which the Rabi frequency, Ω , sets the characteristic timescale of

the system, while the detuning is allowed to vary in time. Specifically, the detuning is updated at discrete time intervals t chosen as

$$t \in [0.01, 0.1, 1, 10, 100] \times \frac{1}{\Omega}. \quad (3)$$

At each time interval listed in Eq. 3, the detuning is randomly drawn from the range

$$\delta \in [-15, 15],$$

and the dynamics are simulated for different Rabi frequencies,

$$\Omega \in [1, 5, 10, 50, 100, 500].$$

Our results show that, even in the presence of time-dependent detuning fluctuations, the two-level system becomes increasingly robust to detuning jitter as the drive strength is increased.

Rabi oscillations with time-dependent detuning $\delta(t)$

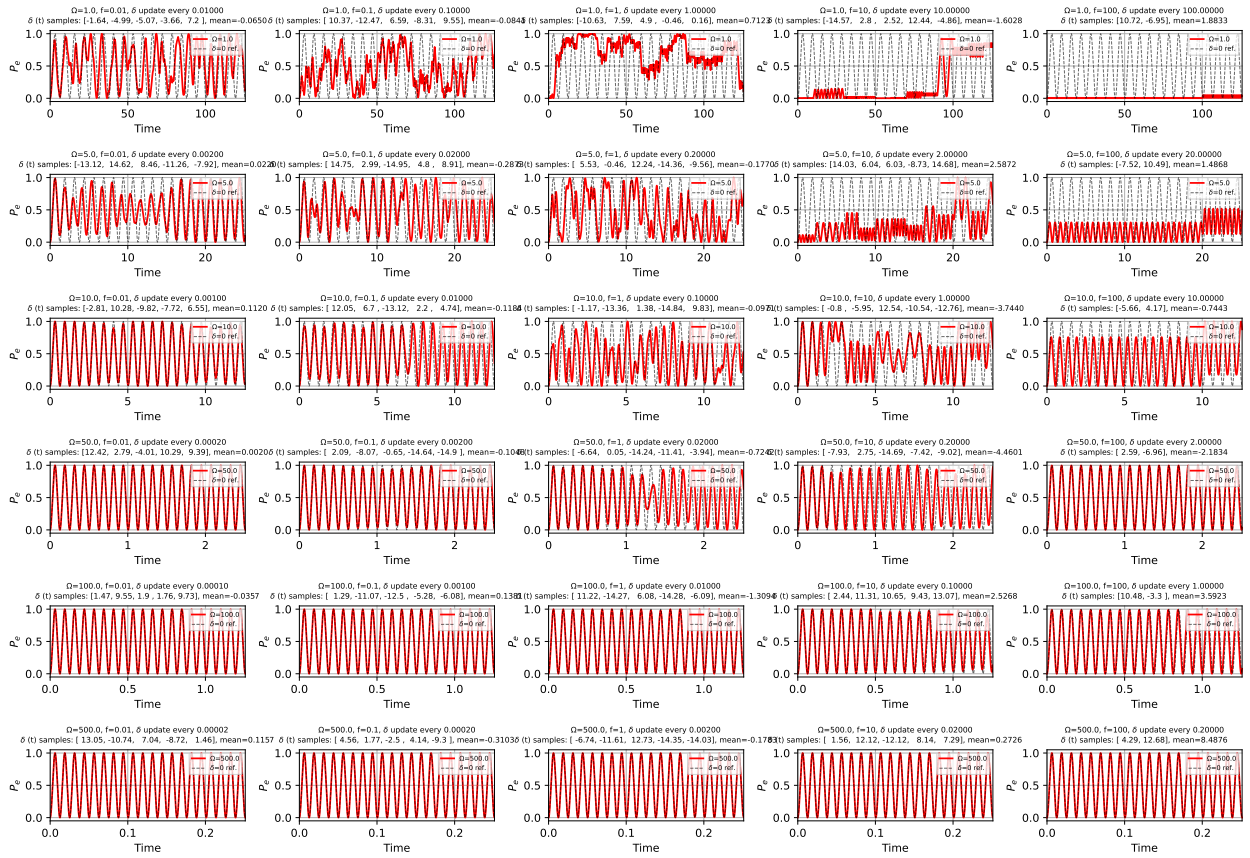


Figure 5: Time Dependent Detuning Simulations

3.4 Rabi Oscillations and Rabi Frequency Jitters

Next, we examine how a two-level system can be made robust against fluctuations in the Rabi frequency,

$$\Omega \rightarrow \Omega \pm \Delta\Omega(t).$$

As suggested in Ref. [3], this can be achieved by applying an additional driving field and moving to a *double-dressed* state basis, which suppresses the effect of Rabi-frequency jitters.

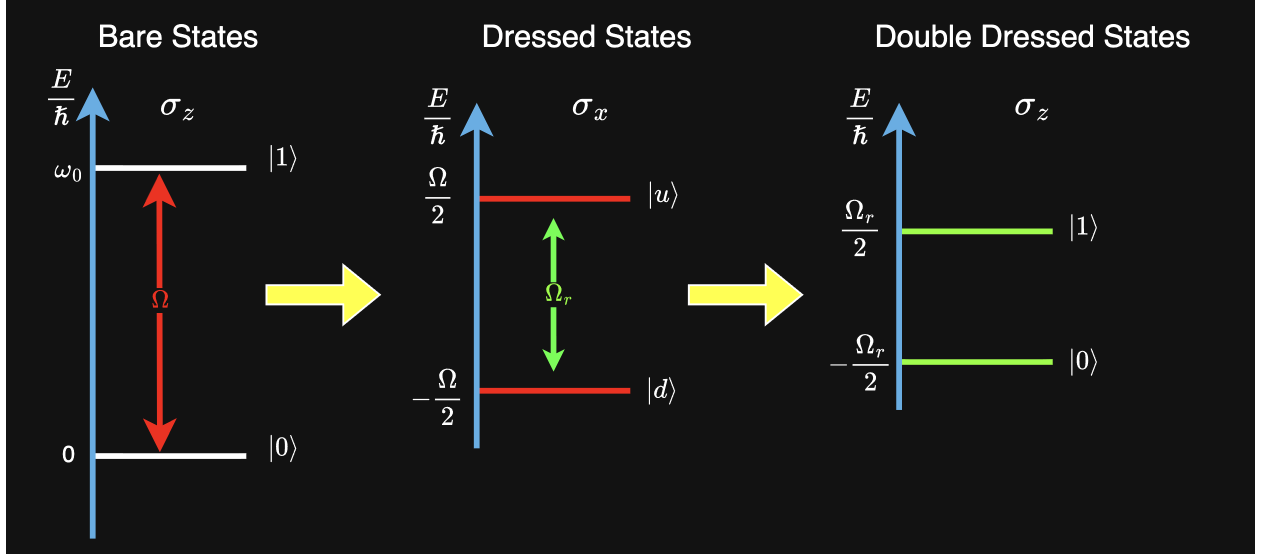


Figure 6: Double Dressed States [3]

Starting from the dressed-state picture associated with a resonant drive of strength Ω_r , an additional driving field is applied on resonance with the dressed-state energy splitting. This second driving field defines a new quantization axis perpendicular to the original dressed-state basis (associated with σ_x), thereby transforming the system into the double-dressed state basis (associated with σ_z). The resulting energy gap along this perpendicular direction protects the system against noise originating from fluctuations in the Rabi frequency.

To illustrate the effect of Rabi-frequency jitter, we perform numerical simulations demonstrating how fluctuations in the drive amplitude lead to a decay of Rabi oscillations. The simulations are carried out for a nominal Rabi frequency $\Omega = 50$, while the amplitude noise strength is varied over the range

$$\Delta\Omega \in [0.05, 0.5, 5].$$

For all simulations, realistic experimental conditions are assumed. The characteristic timescale of the amplitude jitter is chosen relative to the Rabi period, with evolution times taken from the set. The results shown below are averaged for 20 runs.

$$t \in [0.01, 0.1, 1, 10, 100] \times \frac{1}{\Omega}. \quad (4)$$

Averaged Rabi Oscillations with Time-Dependent $\Omega(t)$ jitter

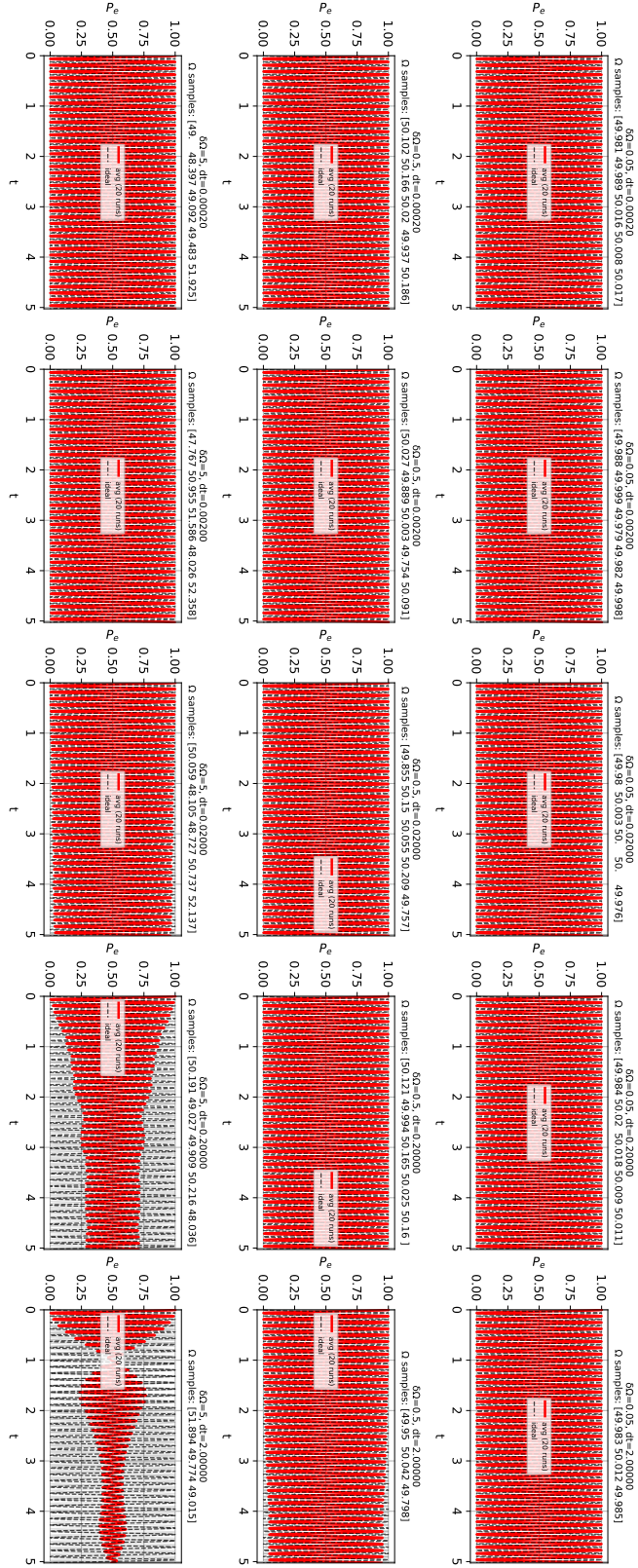


Figure 7: Decay in Rabi Oscillations due to Rabi Frequency Jitters

From the 7 it is evidently seen that the rabi oscillations decay due to larger values of $\Delta\Omega$, which refers to the fact that jitters in the rabi frequency can also lead to system decoherence. In the next section we will try to tackle this with an double dressed state.

3.5 Rabi Oscillations with Phase Modulated Field

We now arrive at a point where we can analyze how *double dressed states* can be used to render a two-level system robust against fluctuations in the Rabi frequency. Nunnerich *et al.* [4] recently demonstrated that a single phase-modulated driving field can effectively realize a double-dressing scheme, leading to a significant enhancement of coherence.

The interaction Hamiltonian describing this driven system is given by

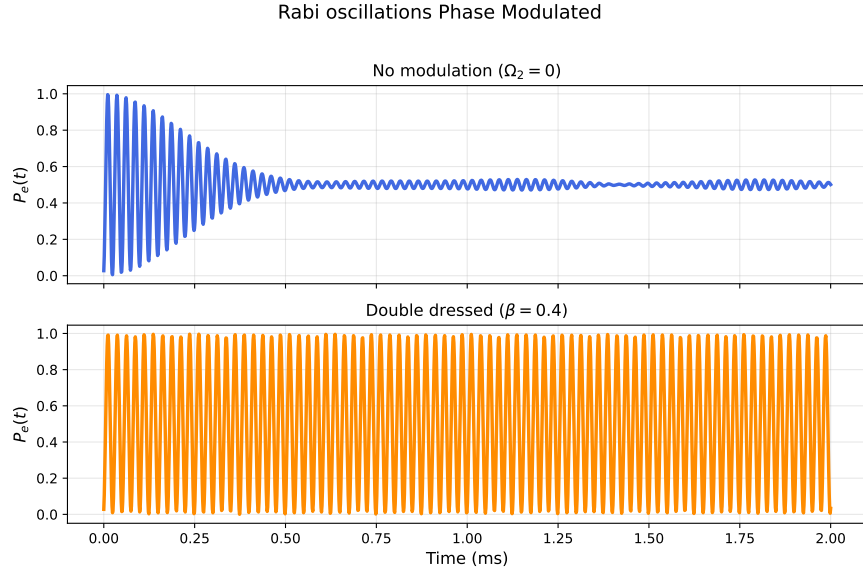
$$H_I(t) = \nu b^\dagger b + \frac{\eta\nu}{2} \sigma_z (b + b^\dagger) + \frac{\Omega_1}{2} \sigma_x - \frac{\Omega_2}{2} \sigma_z \cos(\Omega_1 t), \quad (5)$$

where Ω_1 denotes the Rabi frequency of the resonant radio-frequency drive, and Ω_2 quantifies the amplitude of the phase modulation. The operator σ_z is the Pauli-Z matrix acting on the internal two-level system, ν is the motional mode frequency, and η is the effective Lamb–Dicke parameter. The operators b and $b^\dagger$ are the annihilation and creation operators of the motional mode, respectively.

The phase-modulated drive generates a second effective dressing of the system, such that fluctuations in the primary Rabi frequency are averaged out in the double-dressed basis, thereby suppressing dephasing and extending the coherence time.

For this report, we focus on the internal two-level dynamics and therefore neglect the motional degrees of freedom by omitting the terms proportional to b and $b^\dagger$. The resulting effective Hamiltonian captures the essential physics of the double-dressing mechanism and forms the basis of the numerical simulations presented in the following sections.

A detailed derivation of the full Hamiltonian and the approximations employed is provided in Appendix A.



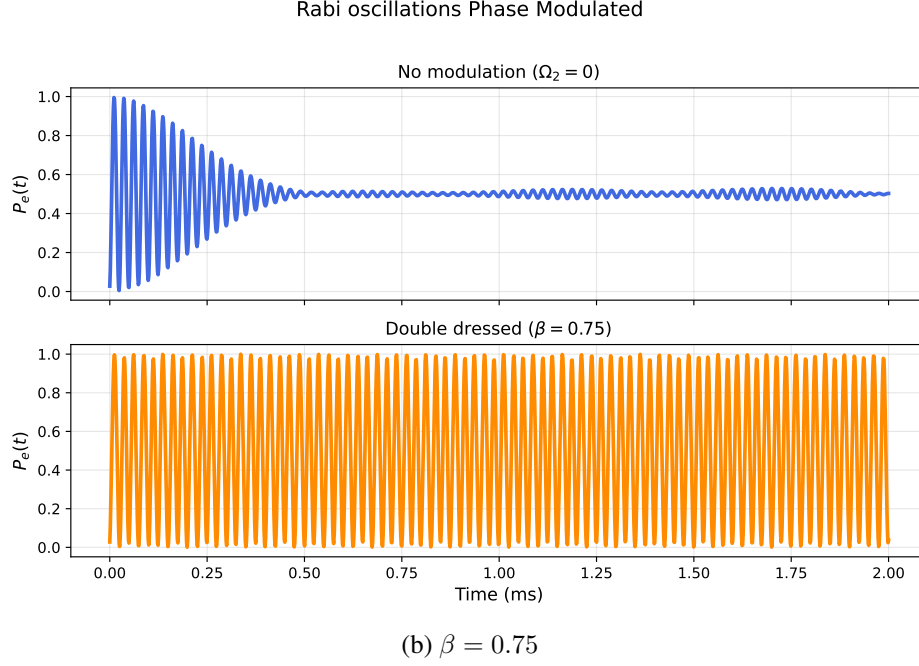


Figure 8: Rabi Oscillations seen with and without Phase modulated drive.

Clearly seen from the 8a that the use of phase modulated drive can indeed improve the coherence time of the the ions. We further in the next section show experimental data taken from the [Ion trap Lab](#), in Aarhus University which helps strongly conclude about the decoupling schemes and its effects

4 Experimental Data

We use the Ion Trap Groups, Ion trapping system to trap Ba^+ ions and use the phase modulation field see a difference in coherence times for the trapped ions.

The parameter β is determined (here used and reffered as m), for a particular modulation frequency Ω_n which is subject to trap frequency as well thus we follow;

$$\Omega_n \left(1 \pm \frac{m}{2} \right) = \underbrace{209.89\text{kHz}}_{\text{Trap Frequency}}$$

The Modulation frequency is chosen as $\Omega_n = 306.372 \text{ kHz}$ for this given Ω_n we have the modulation parameter $m = 0.63506$. Below are the rabi scans performed at different time stamps $[0, 0.5, 60] \text{ ms}$.

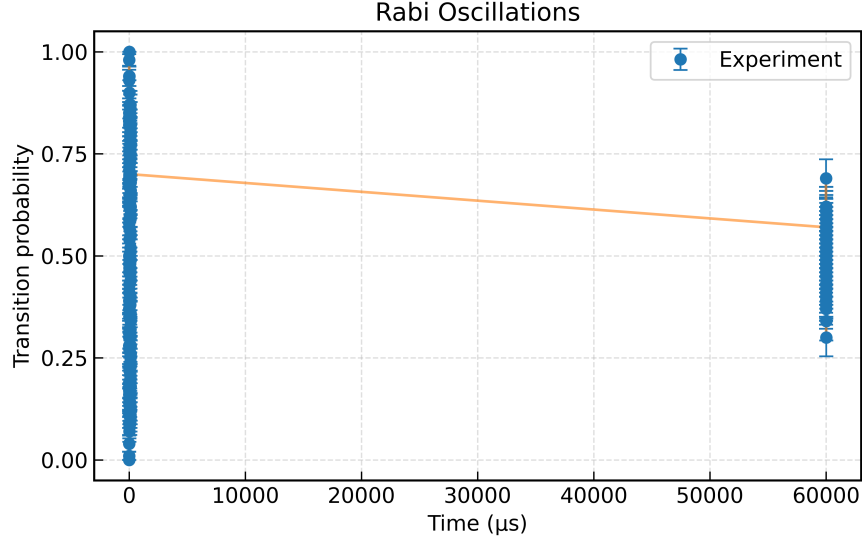


Figure 9: Rabi Scans performed for Phase Modulated Field ($\Omega_n = 306.372 kHz$)

For a easier comparison we also made measurement when $m = 0$, meaning there is no phase modulated field. As one can easily see the field starts to decay out faster in orders of $\sim \mu s$

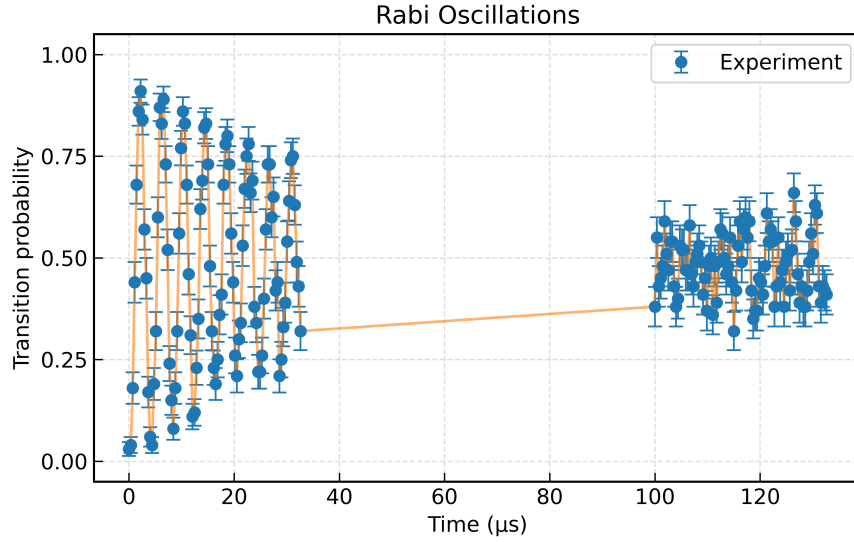


Figure 10: Rabi Scans performed for $m = 0$

From the above experimental data one can confirm the increase in coherence times of ions using a phase modulated fields, which is consistent with the predictions of the simulation results [8a](#) [8](#).

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- [4] Nünnerich, M. et al. Fast, robust, and laser-free universal entangling gates for trapped-ion quantum computing. *Physical Review X* 15, (2025).

A Deriving the Hamiltonian in Interaction Picture

A good amount of math that was skipped between in the Appendix A while going from Eq A6 to A7 in [4]. Lets quickly set up the known variables and what are we trying to acheive with this. All the equations I will refer here in this document are already in [4] if you dont find them let me know.

A.1 Preliminaries

Listing the things we already know;

1. This Equation (A1) describes how the hamiltonian of 2 hyperfine states, given that the ion is subjected to linear magnetic field gradient

$$H = \underbrace{\frac{\omega_0}{2}\sigma_z}_{H_{atom}} + \underbrace{\nu b^\dagger b + \frac{\eta\nu}{2}\sigma_z(b + b^\dagger)}_{H_{motion}} \quad (6)$$

where ω_0 represents the transition frequency of the hyperfine states, σ_z is the Pauli Z matrix, μ is the motional mode frequency, η is the effective Lambe-Dicke parameters, the b , $b^\dagger$ are the lowering and the raising operators respectively. Further in the derivation H can be addressed in its constituent parts H_{atom} & H_{motion}

2. The ion is driven with a phase modulated field which is given by the hamiltonian (A6);

$$H_D(t) = \Omega_1 \sigma_x \cos(\omega_0 t + \phi(t)) \quad (7)$$

where;

$$\phi(t) = \beta \sin(\Omega_1 t), \quad \beta \equiv \frac{\Omega_2}{\Omega_1}, \quad 0 \leq \beta \leq 1$$

Ω_1 is the rabi frequency of the rf drive and Ω_2 is the parameter that quantifies the phase modulation amplitude.

3. The total Hamiltonian $H_t = H + H_D(t)$ is being moved to an Rotating Wave Approximation (RWA) with respect to;

$$H_0(t) = \underbrace{\frac{\omega_0}{2}\sigma_z}_{H_{atom}} + \underbrace{\frac{\Omega_2}{2}\sigma_z \cos(\Omega_1 t)}_{H_b} \quad (8)$$

The point of this report is to use these 3 knowns and end up at the equation A7 in [4]. Before we start the derivation let us have some further equations which can be useful during the derivation.

1. The corresponding unitary $U_0(t)$ to the rotation frame $H_0(t)$ is;

$$U_0(t) = \exp\left[-i \int_0^t H_0(t') dt'\right] = e^{-i \frac{\omega_0 t}{2} \sigma_z} e^{-i \frac{\Omega_2}{2\Omega_1} \sin(\Omega_1 t) \sigma_z}, \quad (9)$$

2. The corresponding unitary $U_0^\dagger(t)$ is;

$$U_0^\dagger(t) = \exp\left[i \int_0^t H_0(t') dt'\right] = e^{i \frac{\omega_0 t}{2} \sigma_z} e^{i \frac{\Omega_2}{2\Omega_1} \sin(\Omega_1 t) \sigma_z}, \quad (10)$$

A.2 Derivation

We are trying to get the hamiltonian H_t in the interaction picture, so the obvious place to start is the interaction picture. Where the $U_0(t)$ is the generator for the hamitonian $H_0(t)$ (think of it is as the generator for the RWA) gives the state;

$$|\psi_I(t)\rangle = U_0^\dagger(t), |\psi(t)\rangle. \quad (11)$$

Starting from the Schrodinger equation in the laboratory frame,

$$i \frac{d}{dt} |\psi(t)\rangle = H_t(t) |\psi(t)\rangle, \quad (12)$$

we differentiate $|\psi_I(t)\rangle$:

$$\frac{d}{dt} |\psi_I(t)\rangle = \dot{U}_0^\dagger(t) |\psi(t)\rangle + U_0^\dagger(t) \frac{d}{dt} |\psi(t)\rangle. \quad (13)$$

Substituting the Schodinger equation and multiplying by i gives

$$i \frac{d}{dt} |\psi_I(t)\rangle = \left[U_0^\dagger(t) H_t(t) U_0(t) + i \dot{U}_0^\dagger(t) U_0(t) \right] |\psi_I(t)\rangle. \quad (14)$$

Hence the interaction-picture Hamiltonian is

$$H_I(t) = U_0^\dagger(t) H_t(t) U_0(t) + i \dot{U}_0^\dagger(t) U_0(t). \quad (15)$$

Using $U_0^\dagger U_0 = \mathbb{I}$ and differentiating yields

$$\dot{U}_0^\dagger U_0 = -U_0^\dagger \dot{U}_0, \quad (16)$$

so that

$$H_I(t) = U_0^\dagger(t) H_t(t) U_0(t) - i U_0^\dagger(t) \dot{U}_0(t). \quad (17)$$

By construction, $U_0(t)$ satisfies

$$i \dot{U}_0(t) = H_0(t) U_0(t), \quad (18)$$

which implies

$$i U_0^\dagger(t) \dot{U}_0(t) = U_0^\dagger(t) H_0(t) U_0(t). \quad (19)$$

And therefore the interaction picture hamiltonian becomes;

$$\boxed{H_I(t) = U_0^\dagger(t) [H_t(t) - H_0(t)] U_0(t)} \quad (20)$$

where $H_t(t)$ is the full hamiltonian (system + drive) and H_0 is the RWA Frame

We further expliot the equation 20. We expand using 6 and 8 and solve each one of them.

$$H_I(t) = U_0^\dagger(t) [H_t(t) - H_0(t)] U_0(t) \quad (21)$$

becomes;

$$H_I(t) = U_0^\dagger(t) [\cancel{H_{atom}} + H_{motion} + H_D(t) - \cancel{H_{atom}} - H_b(t)] U_0(t) \quad (22)$$

Finally we have

$$H_I(t) = U_0^\dagger(t) H_{motion} U_0(t) + U_0^\dagger(t) H_D(t) U_0(t) - U_0^\dagger(t) H_b(t) U_0(t) \quad (23)$$

Dealing with all of these parts one by tackling the first term $U_0^\dagger(t) H_{\text{motion}} U_0(t)$

$$U_0^\dagger(t) H_{\text{motion}} U_0(t) = U_0^\dagger(t) (\nu b^\dagger b + \frac{\eta\nu}{2} \sigma_z (b + b^\dagger)) U_0(t) = \nu b^\dagger b + \frac{\eta\nu}{2} \sigma_z (b + b^\dagger)$$

Since the term $\nu b^\dagger b$ refers to the motional states and the unitary $U_0(t)$ acts only on the internal spin states of the ion, we see no change in this terms (different hilbert spaces). $[U_0(t), \nu b^\dagger b] = 0$. In the interaction picture when we have for any operator A; that commutes $[U_0(t), A] = 0$ it remains unchanged, $U_0^\dagger(t) A U_0(t) = A$

$$U_0^\dagger(t) A U_0(t) = U_0^\dagger(t) U_0(t) A = \mathcal{I} A = A$$

Similar argument can be used for the term with σ_z here. The unitary $U_0(t)$ is made up of σ_z (same generator) therefore these terms also commute $[U_0(t), \sigma_z] = 0$

Therefore;

$$U_0^\dagger(t) H_{\text{motion}} U_0(t) = \nu b^\dagger b + \frac{\eta\nu}{2} \sigma_z (b + b^\dagger) \quad (24)$$

Tackling the second term; $U_0^\dagger(t) H_D(t) U_0(t)$

we remember how 9 and 10 look like. we will expand the $\cos$ term in the $H_D(t)$ in the exponentials and write the σ_x as sum of ladder operators;

$$\cos(\omega t + \phi(t)) = \frac{e^{i(\omega t + \phi(t))} + e^{-i(\omega t + \phi(t))}}{2}$$

with

$$\sigma_x = \sigma_+ + \sigma_-$$

Therefore 7 becomes;

$$\begin{aligned} H_D(t) &= \Omega_1 (\sigma_+ + \sigma_-) \left(\frac{e^{i(\omega t + \phi(t))} + e^{-i(\omega t + \phi(t))}}{2} \right) \\ &= \frac{\Omega_1}{2} \left(e^{i(\omega t + \phi(t))} \sigma_+ + e^{i(\omega t + \phi(t))} \sigma_- + e^{-i(\omega t + \phi(t))} \sigma_+ + e^{-i(\omega t + \phi(t))} \sigma_- \right) \end{aligned} \quad (25)$$

Finally solving for required term we get;

$$U_0^\dagger(t) H_D(t) U_0(t) = \frac{\Omega_1}{2} U_0^\dagger(t) \left(e^{i(\omega t + \phi(t))} \sigma_+ + e^{i(\omega t + \phi(t))} \sigma_- + e^{-i(\omega t + \phi(t))} \sigma_+ + e^{-i(\omega t + \phi(t))} \sigma_- \right) U_0(t) \quad (26)$$

$$\begin{aligned} &= \frac{\Omega_1}{2} \left(U_0^\dagger(t) e^{i(\omega t + \phi(t))} \sigma_+ U_0(t) \right) + \left(U_0^\dagger(t) e^{i(\omega t + \phi(t))} \sigma_- U_0(t) \right) \\ &\quad + \left(U_0^\dagger(t) e^{-i(\omega t + \phi(t))} \sigma_+ U_0(t) \right) + \left(U_0^\dagger(t) e^{-i(\omega t + \phi(t))} \sigma_- U_0(t) \right) \end{aligned} \quad (27)$$

Using the relation, (for proof check appendix B)

$$U_0^\dagger(t) \sigma_\pm U_0(t) = e^{\pm i\omega_0 t} \sigma_\pm,$$

$$\begin{aligned}
&= \frac{\Omega_1}{2} \left(e^{i\omega t} e^{\phi(t)} U_0^\dagger(t) \sigma_+ U_0(t) \right) + \left(e^{i\omega t} e^{\phi(t)} U_0^\dagger(t) \sigma_- U_0(t) \right) \\
&\quad + \left(e^{-i\omega t} e^{-\phi(t)} U_0^\dagger(t) \sigma_+ U_0(t) \right) + \left(e^{-i\omega t} e^{-\phi(t)} U_0^\dagger(t) \sigma_- U_0(t) \right)
\end{aligned}$$

finally giving;

$$= \frac{\Omega_1}{2} \left(e^{i\omega t} e^{\phi(t)} e^{i\omega t} \sigma_+ \right) + \left(e^{i\omega t} e^{\phi(t)} e^{-i\omega t} \sigma_- \right) + \left(e^{-i\omega t} e^{-\phi(t)} e^{i\omega t} \sigma_+ \right) + \left(e^{-i\omega t} e^{-\phi(t)} e^{-i\omega t} \sigma_- \right) \quad (28)$$

$$= \frac{\Omega_1}{2} \underbrace{\left(e^{2i\omega t} e^{\phi(t)} \sigma_+ \right)}_{RWA} + \left(e^{i\omega t} e^{\phi(t)} e^{-i\omega t} \sigma_- \right) + \left(e^{-i\omega t} e^{-\phi(t)} e^{i\omega t} \sigma_+ \right) + \underbrace{\left(e^{-2i\omega t} e^{-\phi(t)} \sigma_- \right)}_{RWA} \quad (29)$$

$$= \frac{\Omega_1}{2} (\sigma_+ + \sigma_-) \quad (30)$$

$$= \frac{\Omega_1}{2} \sigma_x \quad (31)$$

Tackling the final term $-U_0^\dagger(t) H_b(t) U_0(t)$

$$-U_0^\dagger(t) H_b(t) U_0(t) = -U_0^\dagger(t) \frac{\Omega_2}{2} \sigma_z \cos(\Omega_1 t) U_0(t) \quad (32)$$

The unitary $U_0(t)$ is made up of σ_z (same generator) therefore these terms also commute $[U_0(t), \sigma_z] = 0$. Therefore we have,

$$-U_0^\dagger(t) H_b(t) U_0(t) = -\frac{\Omega_2}{2} \sigma_z \cos(\Omega_1 t) \quad (33)$$

Concluding, we finally arrive to the solution of the 23 combining 24, 31 and 33 we have;

$$\boxed{H_I(t) = \nu b^\dagger b + \frac{\eta\nu}{2} \sigma_z (b + b^\dagger) + \frac{\Omega_1}{2} \sigma_x - \frac{\Omega_2}{2} \sigma_z \cos(\Omega_1 t)} \quad (34)$$

which is exactly the equation shown in [4] as equation A7.

B Important Relation

This section presents full derivations of the identity

$$U_0^\dagger(t) \sigma_\pm U_0(t) = e^{\pm i\omega_0 t} \sigma_\pm, \quad (35)$$

Here we assume that

$$U_0(t) = \underbrace{e^{-i\frac{\omega_0 t}{2}\sigma_z}}_{FastRotation} \underbrace{e^{-i\frac{\Omega_2}{2\Omega_1}\sin(\Omega_1 t)\sigma_z}}_{SlowRotation}$$

and work only with the faster term. Since the slower oscillation term ($\Omega_1, \Omega_2 \ll \omega_0$) is a engineered term it has no information about the dynamics of ω_0 . Moreover if really needed the dynamics of this term can be considered at another step as a different unitary.

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (36)$$

$$U_0(t) = e^{-i\frac{\omega_0 t}{2}\sigma_z} = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{+i\omega_0 t/2} \end{pmatrix}, \quad (37)$$

$$U_0^\dagger(t) = \begin{pmatrix} e^{+i\omega_0 t/2} & 0 \\ 0 & e^{-i\omega_0 t/2} \end{pmatrix}. \quad (38)$$

First multiply on the right:

$$\sigma_+ U_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{+i\omega_0 t/2} \end{pmatrix} = \begin{pmatrix} 0 & e^{+i\omega_0 t/2} \\ 0 & 0 \end{pmatrix}. \quad (39)$$

Then multiply on the left:

$$U_0^\dagger(\sigma_+ U_0) = \begin{pmatrix} e^{+i\omega_0 t/2} & 0 \\ 0 & e^{-i\omega_0 t/2} \end{pmatrix} \begin{pmatrix} 0 & e^{+i\omega_0 t/2} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & e^{+i\omega_0 t} \\ 0 & 0 \end{pmatrix}. \quad (40)$$

Therefore,

$$\boxed{U_0^\dagger \sigma_+ U_0 = e^{+i\omega_0 t} \sigma_+}. \quad (41)$$

Repeating the same steps gives

$$\boxed{U_0^\dagger \sigma_- U_0 = e^{-i\omega_0 t} \sigma_-}. \quad (42)$$